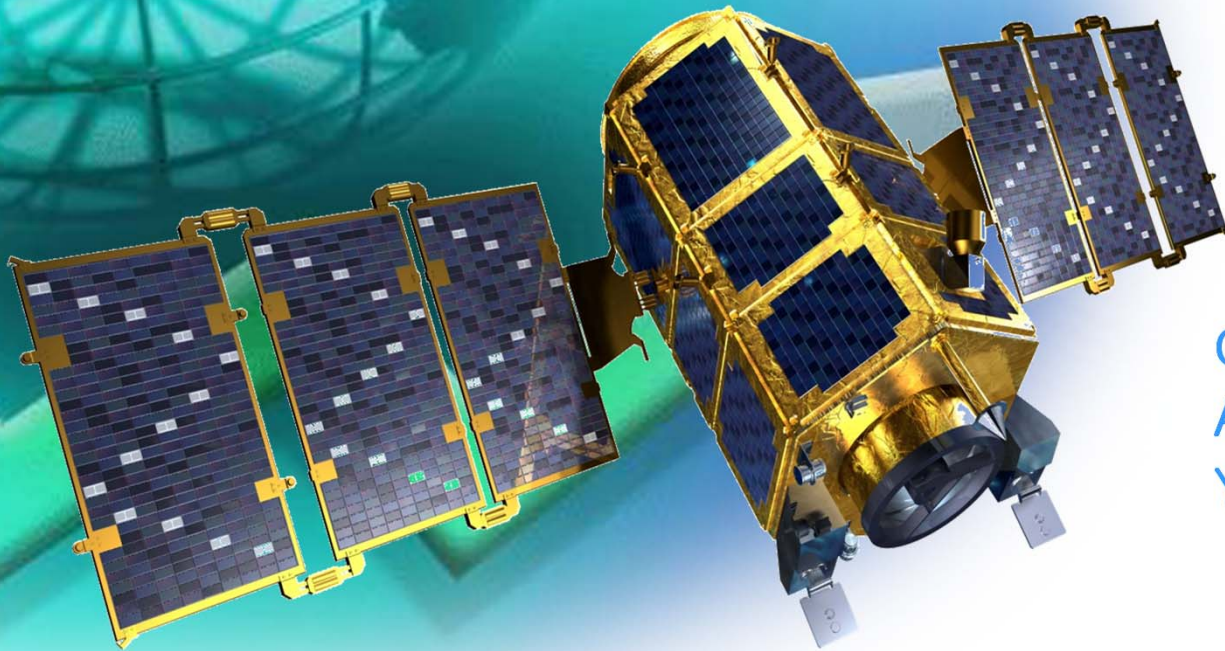
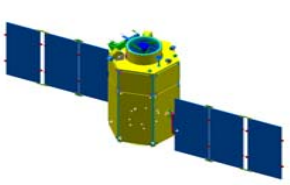


Fall 2011

# ***Spacecraft Systems***



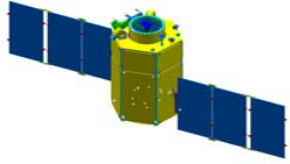
Chandeok Park  
Astrodynamics & Control Lab.  
Yonsei University



# ***Chapter 11***

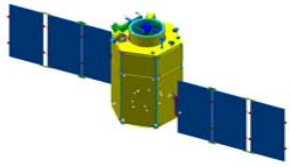
## ***SPACECRAFT SUBSYSTEMS***

- 11.1 Attitude Determination & Control Subsystem
- 11.2 Telemetry, Tracking & Command Subsystem
- 11.3 Command & Data Handling Subsystem
- 11.4 (Electrical) Power Subsystem
- 11.5 Thermal Control Subsystem
- 11.6 Structures & Mechanism Subsystem**
- 11.7 Guidance, Navigation & Control Subsystem

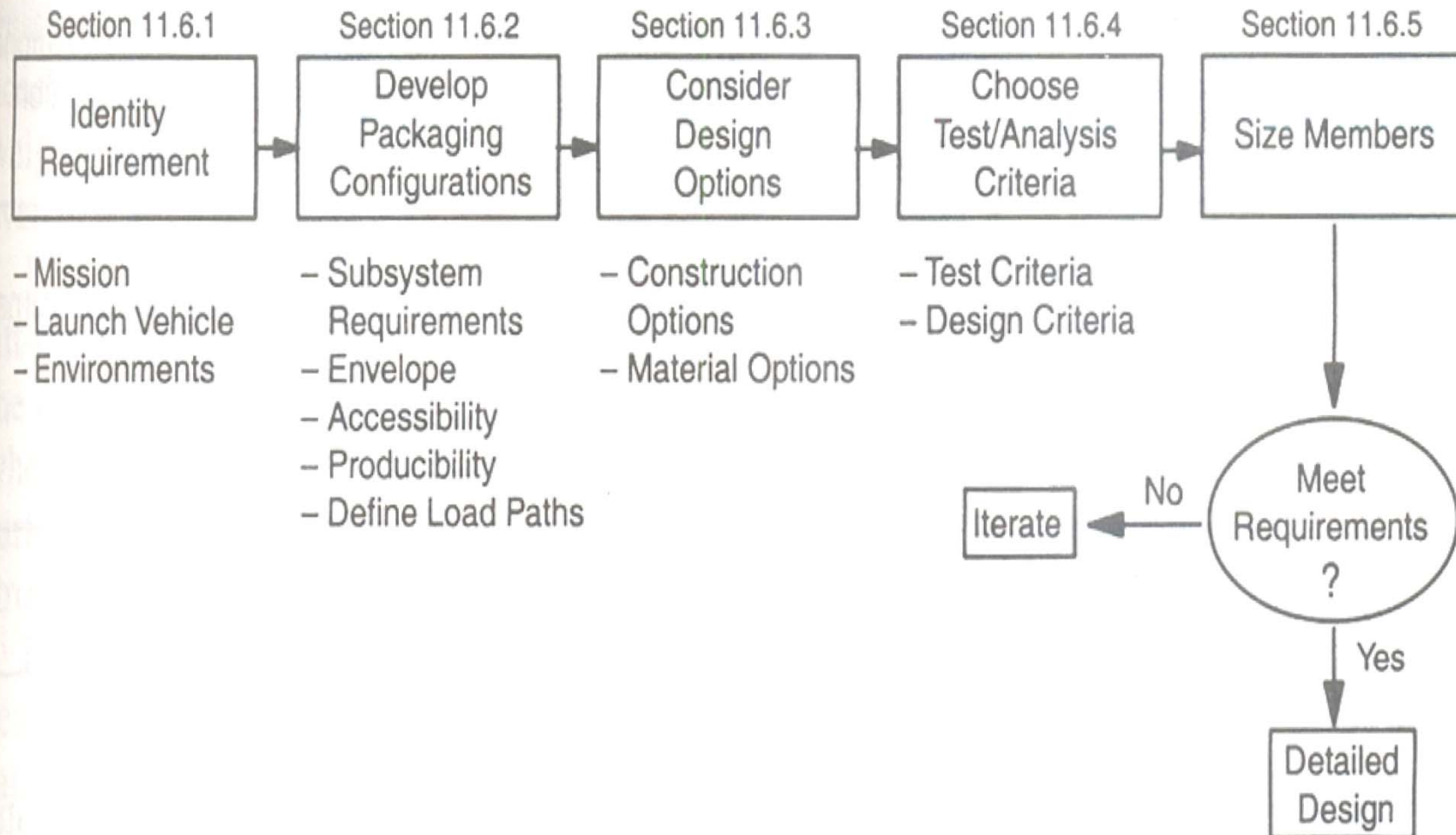


## ***11.6 Structures and Mechanisms***

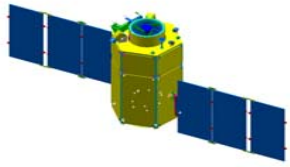
- ◆ Structures and Mechanisms Subsystem
  - Mechanically support all other spacecraft subsystems
  - Attaches S/C to the launch vehicle
  - Primary Structure carries the S/C's major loads
  - Secondary Structure supports:
    - wire bundles
    - propellant lines
    - nonstructural doors
    - brackets for components typically under 5 kg.
- ◆ Design must satisfy all strength and stiffness requirements of the S/C and of its interface to the booster



# 11.6 Structures and Mechanisms



**Fig. 11-25. The Preliminary Design Process for Structures and Mechanisms.** We move from left to right, iterating as needed, when designing the spacecraft structure.



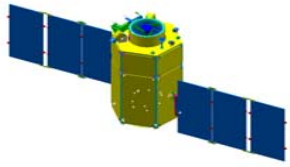
# 11.6 Structures and Mechanisms

## ◆ 11.6.1 Structural Requirements

- Launch Vehicle is the most obvious source of structural requirements, dictating the S/C weight, geometry, rigidity, and strength
- In actual design, we combine math models of the S/C and launch vehicle to do a coupled loads analysis.

**TABLE 11-50. Typical Sources for Structural Requirements by Mission Phase.** The structural design must account for specific loads in every phase.

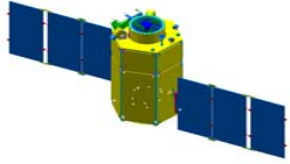
| Mission Phase                   | Source of Requirements                                                                                                                                    |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Manufacture and Assembly</i> | <ul style="list-style-type: none"><li>• Handling fixture or container reactions</li><li>• Stresses induced by manufacturing processes (welding)</li></ul> |
| <i>Transport and Handling</i>   | <ul style="list-style-type: none"><li>• Crane or dolly reactions</li><li>• Land, sea, or air transport environments</li></ul>                             |
| <i>Testing</i>                  | <ul style="list-style-type: none"><li>• Environments from vibration or acoustic tests</li><li>• Test fixture reaction loads</li></ul>                     |
| <i>Prelaunch</i>                | <ul style="list-style-type: none"><li>• Handling during stacking sequence and pre-flight checks</li></ul>                                                 |



# 11.6 Structures and Mechanisms

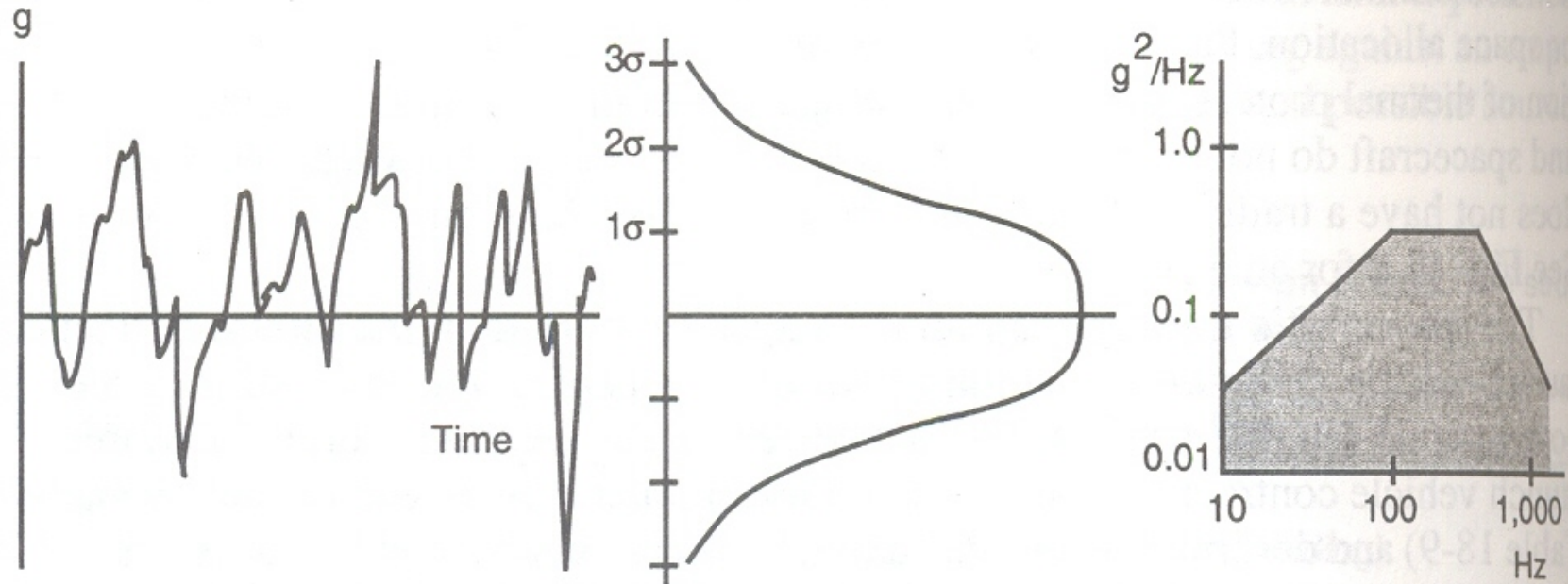
## ◆ 11.6.1 Structural Requirements (cont'd)

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Launch and Ascent</i>                   | <ul style="list-style-type: none"><li>• Steady-state booster accelerations</li><li>• Vibro-acoustic noise during launch and transonic phase</li><li>• Propulsion system engine vibrations</li><li>• Transient loads during booster ignition and burn-out, stage separations, vehicle maneuvers, propellant slosh, and payload fairing separation</li><li>• Pyrotechnic shock from separation events</li></ul> |
| <i>Mission Operations</i>                  | <ul style="list-style-type: none"><li>• Steady-state thruster accelerations</li><li>• Transient loads during pointing maneuvers and attitude control burns or docking events</li><li>• Pyrotechnic shock from separation events, deployments</li><li>• Thermal environments</li></ul>                                                                                                                         |
| <i>Reentry and Landing (if applicable)</i> | <ul style="list-style-type: none"><li>• Aerodynamic heating</li><li>• Transient wind and landing loads</li></ul>                                                                                                                                                                                                                                                                                              |

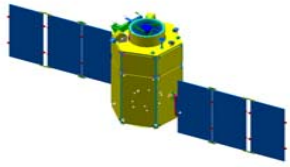


# 11.6 Structures and Mechanisms

## ◆ 11.6.1 Structural Requirements (cont'd)



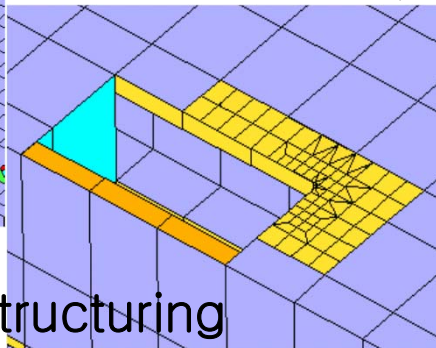
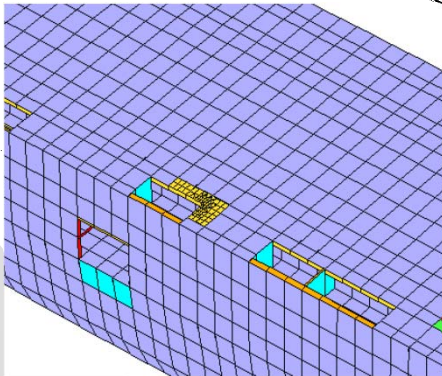
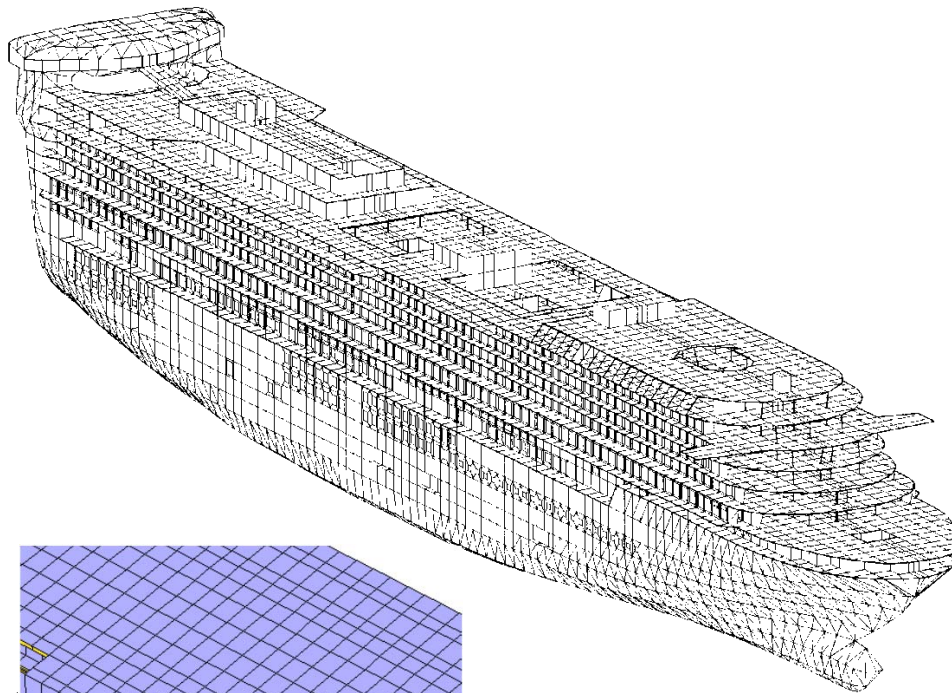
**Fig. 11-26. Random Vibration.** The plot shown on the left is an example of how acceleration from random vibration would vary over time. The probability density curve (center), with the same vertical scale, describes the relative likelihood of acceleration being at a given value. This is a normal distribution, with each tick mark on the vertical scale representing a standard deviation,  $\sigma$ . The figure on the right is a plot of power spectral density. It describes the frequency content of the vibration and is equal to the mean-square acceleration ( $g^2$ ) in a selected frequency band divided by the width, in Hz, of that band.



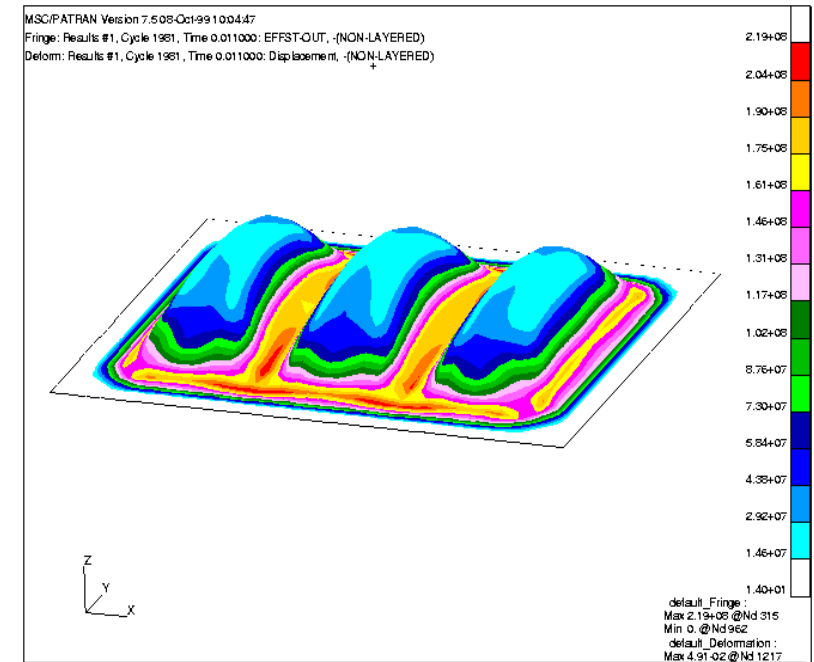
# 11.6 Structures and Mechanisms

## ◆ 11.6.1 Structural Requirements (cont'd)

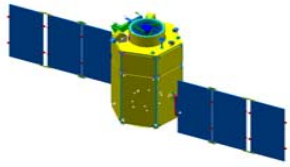
- Finite Element Method



Example of Substructuring



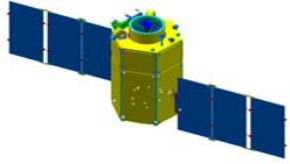
Effective Stress Distributions  
(units : Pa)



# ***11.6 Structures and Mechanisms***

## ◆ 11.6.2 Packing & Configuring the Subsystem

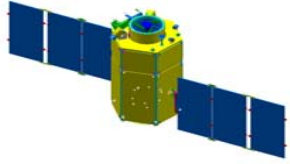
- Designers must trade the low weight of a high-density design against the need to access individual components for testing or replacement before launch
- Sensing devices always require specific fields of view and pointing accuracy
- Communication antennas require rigidity, thermo-elastic stability, and a clear field of view
- Components for C&DH are placed in the center of S/C to shield against radiation
- Thrusters must be kept from contaminating sensors, antennas, and solar array cells with propellant gases
  - Thrust vector should be aligned with S/C center of mass
- Batteries should be accessible for pre-launch testing or replacement, and placed where they will be at their optimum temperature
- Thermal control specialists can place components, select materials, and suggest surrounding structure to help control temperature



# 11.6 Structures and Mechanisms

## ◆ 11.6.3 Design Options

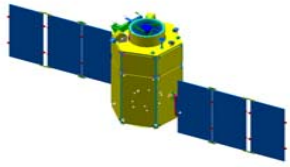
- In designing a structure, we consider optional materials, types of structure, and methods of construction
  - To select from these options, we do trade studies to compare weight, cost, and risk
- Materials are selected based on:
  - Strength & Stiffness
  - Density (Weight)
  - Thermal Conductivity & Thermal Expansion
  - Corrosion Resistance
  - Cost
  - Ductility (which can prevent cracks)
  - Fracture Toughness (ability to resist crack growth)
  - Ease of Fabrication
  - Versatility of Attachment Options
  - Availability



# 11.6 Structures and Mechanisms

## ◆ 11.6.3 Design Options (cont'd)

- Metallic Materials
  - Homogeneous & Isotropic
  - Aluminum Alloy is the most commonly used metal for S/C structure
    - Relatively lightweight, strong, readily available, easy to machine, and low cost
  - Steel or Titanium are normally chosen for harder or denser materials
- Nonmetallic Materials: Composites
  - Not Homogeneous, Not Isotropic
  - Graphite–Epoxy is one popular advanced composite
    - Has graphite fibers for strength and stiffness in an epoxy matrix

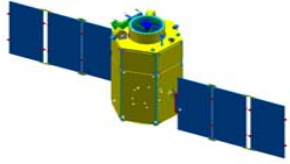


# 11.6 Structures and Mechanisms

## ◆ 11.6.3 Design Options (cont'd)

TABLE 11-51. Advantages and Disadvantages of Commonly Used Materials.

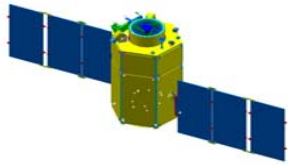
| Material                     | Advantages                                                                                                                                                                                                | Disadvantages                                                                                                                                               |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Aluminum</i>              | <ul style="list-style-type: none"><li>• High strength vs. weight</li><li>• Ductile; tolerant of concentrated stresses</li><li>• Easy to machine</li><li>• Low density; efficient in compression</li></ul> | <ul style="list-style-type: none"><li>• Relatively low strength vs. volume</li><li>• Low hardness</li><li>• High coefficient of thermal expansion</li></ul> |
| <i>Steel</i>                 | <ul style="list-style-type: none"><li>• High strength</li><li>• Wide range of strength, hardness, and ductility obtained by treatment</li></ul>                                                           | <ul style="list-style-type: none"><li>• Not efficient for stability (high density)</li><li>• Most are hard to machine</li><li>• Magnetic</li></ul>          |
| <i>Heat-resistant alloys</i> | <ul style="list-style-type: none"><li>• High strength vs. volume</li><li>• Strength retained at high temperatures</li><li>• Ductile</li></ul>                                                             | <ul style="list-style-type: none"><li>• Not efficient for stability (high density)</li><li>• Not as hard as some steels</li></ul>                           |
| <i>Magnesium</i>             | <ul style="list-style-type: none"><li>• Low density—very efficient for stability</li></ul>                                                                                                                | <ul style="list-style-type: none"><li>• Susceptible to corrosion</li><li>• Low strength vs. volume</li></ul>                                                |



# 11.6 Structures and Mechanisms

## ◆ 11.6.3 Design Options (cont'd)

|                  |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Titanium</i>  | <ul style="list-style-type: none"><li>• High strength vs. weight</li><li>• Low coefficient of thermal expansion</li></ul>                                                                                                         | <ul style="list-style-type: none"><li>• Hard to machine</li><li>• Poor fracture toughness if solution treated and aged</li></ul>                                                                                                                                                                               |
| <i>Beryllium</i> | <ul style="list-style-type: none"><li>• High stiffness vs. density</li></ul>                                                                                                                                                      | <ul style="list-style-type: none"><li>• Low ductility &amp; fracture toughness</li><li>• Low short transverse properties</li><li>• Toxic</li></ul>                                                                                                                                                             |
| <i>Composite</i> | <ul style="list-style-type: none"><li>• Can be tailored for high stiffness, high strength, and extremely low coefficient of thermal expansion</li><li>• Low density</li><li>• Good in tension (e.g., pressurized tanks)</li></ul> | <ul style="list-style-type: none"><li>• Costly for low production volume; requires development program</li><li>• Strength depends on workmanship; usually requires individual proof testing</li><li>• Laminated composites are not as strong in compression</li><li>• Brittle; can be hard to attach</li></ul> |

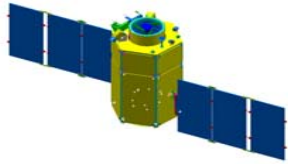


# 11.6 Structures and Mechanisms

## ◆ 11.6.3 Design Options (cont'd)

**TABLE 11-52. Design Properties for Commonly Used Metals [MIL-HDBK-5G, 1994].** The design allowable stresses given here are statistically guaranteed for at least 99% of all material specimens. Strengths shown are for the *longitudinal* direction (rolling or extrusion direction) of the material; strengths are usually lower in the *long-transverse* (across width) and *short-transverse* (through thickness) directions.

| Material Alloy and Form | $\rho$<br>10 <sup>3</sup> kg/m <sup>3</sup><br>(lb/in <sup>3</sup> ) | $F_{tu}$<br>10 <sup>6</sup> N/m <sup>2</sup><br>(10 <sup>3</sup> lb/in <sup>2</sup> ) | $F_{cy}$<br>10 <sup>6</sup> N/m <sup>2</sup><br>(10 <sup>3</sup> lb/in <sup>2</sup> ) | $E$<br>10 <sup>9</sup> N/m <sup>2</sup><br>(10 <sup>6</sup> lb/in <sup>2</sup> ) | $e$<br>% | $\alpha$<br>10 <sup>-6</sup> /°C<br>(10 <sup>-6</sup> /°F) |
|-------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------|------------------------------------------------------------|
| <i>Aluminum</i>         |                                                                      |                                                                                       |                                                                                       |                                                                                  |          |                                                            |
| 2219-T851<br>1" Plate   | 2.85<br>(0.103)                                                      | 420<br>(61)                                                                           | 320<br>(47)                                                                           | 72<br>(10.5)                                                                     | 7        | 22.1<br>(12.3)                                             |
| 6061-T6<br>Bar          | 2.71<br>(0.098)                                                      | 290<br>(42)                                                                           | 240<br>(35)                                                                           | 68<br>(9.9)                                                                      | 10       | 22.9<br>(12.7)                                             |
| 7075-T73<br>Sheet       | 2.80<br>(0.101)                                                      | 460<br>(67)                                                                           | 380<br>(55)                                                                           | 71<br>(10.3)                                                                     | 8        | 22.1<br>(12.3)                                             |
| <i>Steel</i>            |                                                                      |                                                                                       |                                                                                       |                                                                                  |          |                                                            |
| 17-4PH H1150z<br>Bar    | 7.86<br>(0.284)                                                      | 860<br>(125)                                                                          | 620<br>(90)                                                                           | 196<br>(28.5)                                                                    | 16       | 11.2<br>(6.2)                                              |



# 11.6 Structures and Mechanisms

## ◆ 11.6.3 Design Options (cont'd)

|                                                |                 |               |                |               |    |                |
|------------------------------------------------|-----------------|---------------|----------------|---------------|----|----------------|
| <i>Heat-Res. Alloy</i><br>A-286<br>2" Bar      | 7.94<br>(0.287) | 970<br>(140)  | 660<br>(95)    | 201<br>(29.1) | 12 | 16.2<br>(9.0)  |
| Inconel 718<br>4" Bar                          | 8.22<br>(0.297) | 1280<br>(185) | 1,080<br>(156) | 203<br>(29.4) | 12 | 12.2<br>(6.8)  |
| <i>Magnesium</i><br>AZ31B H24<br>Sheet         | 1.77<br>(0.064) | 270<br>(39)   | 165<br>(24)    | 45<br>(6.5)   | 6  | 25.4<br>(14.1) |
| <i>Titanium</i><br>Ti-6Al-4V<br>Annealed Plate | 4.43<br>(0.160) | 900<br>(130)  | 855<br>(124)   | 110<br>(16.0) | 10 | 8.8<br>(4.9)   |
| <i>Beryllium</i><br>AMS 7906<br>Bar            | 1.85<br>(0.067) | 320<br>(47)   | —              | 290<br>(42)   | 2  | 11.5<br>(6.4)  |

$\rho$  = Density

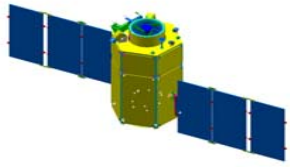
$F_{tu}$  = Allowable Tensile Ultimate Stress, the highest uni-axial tensile stress a material can sustain before rupturing.

$F_{cy}$  = Allowable Compressive Yield Stress, the compressive stress that causes a permanent deformation of 0.2% of the specimen's length.

$E$  = Young's Modulus, a.k.a. Modulus of Elasticity, the ratio of stress to strain (length change divided by original length) in the linear elastic range (see Sec. 11.6.6).

$e$  = Elongation, a measure of ductility, equal to the percentage change in length caused by plastic deformation prior to rupture.

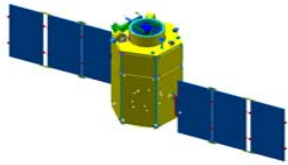
$\alpha$  = Coefficient of Thermal Expansion, a measure of strain per degree temperature change. Values shown for  $\alpha$  are at room temperature.



# ***11.6 Structures and Mechanisms***

## ◆ 11.6.3 Design Options (cont'd)

- Types of Structures include:
  - Skin Panel Assemblies (Monocoque, Semi-monocoque)
  - Trusses
  - Ring Frames
  - Pressure Vessels
  - Fittings
  - Brackets
  - Equipment Boxes
- Method of Construction: Attach structural elements with:
  - Adhesive Bonds
  - Welds
  - Mechanical Fasteners



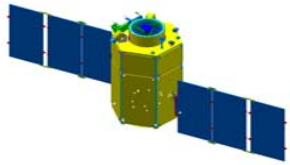
# 11.6 Structures and Mechanisms

## ◆ 11.6.4 Structural Design Philosophy & Criteria

- To develop a structure light enough for flight, and to keep S/C affordable, we must accept risk of failure

**TABLE 11-53. Terms and Criteria Used in Strength Analysis.** We design space structures to meet specified or selected criteria for preventing yield and ultimate failures. A *yield failure* is one in which the structure suffers permanent deformation that degrades the mission; *ultimate failure* is rupture or collapse. Two factors of safety—one for yield and one for ultimate—typically apply to a structural assembly, and depend on the selected test option (Table 11-54). For each structural member, the allowable load, the design load, and the margin of safety each have two values, one for yield and one for ultimate.

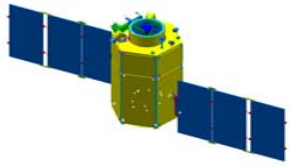
| Term                                                | Definition                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Load Factor</i>                                  | A multiple of weight on Earth, representing the force of inertia that resists acceleration. The load factor applies in the direction opposite that of the acceleration. For example, an object under an acceleration of 5 g, where g is the gravitational acceleration, has a load factor of $-5$ ; if that object weighs 500 N (a mass of 51 kg), the force it exerts on its support structure is $-2,500$ N. |
| <i>Limit Load</i><br>(or <i>design limit load</i> ) | The maximum load expected during the mission or for a given event, at a specified or selected statistical probability (typically 99% for expendable launch vehicles and 99.87% for launches with humans aboard). The load can be acceleration, load factor, force, or moment.                                                                                                                                  |



# 11.6 Structures and Mechanisms

## ◆ 11.6.4 S. D. Philosophy & Criteria (cont'd)

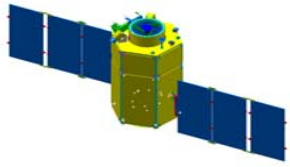
|                                 |                                                                                                                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Allowable Load or Stress</i> | The highest load or stress a structure or material can withstand without failure, based on statistical probability (usually 99%; i.e., only 1% chance the actual strength is less than the allowable). |
| <i>Factor of Safety, FS</i>     | A factor applied to the limit load to obtain the design load for the purpose of decreasing the chance of failure.                                                                                      |
| <i>Design Load</i>              | Limit load multiplied by the yield or ultimate factor of safety; this value must be no greater than the corresponding allowable load.                                                                  |
| <i>Design Stress</i>            | Predicted stress caused by the design load; this value must not exceed the corresponding allowable stress.                                                                                             |
| <i>Margin of Safety, MS</i>     | A measure of reserve strength:<br>$MS = \frac{\text{Allowable load (or stress)}}{\text{Design load (or stress)}} - 1 \geq 0 \text{ to satisfy design criteria}$                                        |



# 11.6 Structures and Mechanisms

## ◆ 11.6.5 Preliminary Sizing of Structural Members

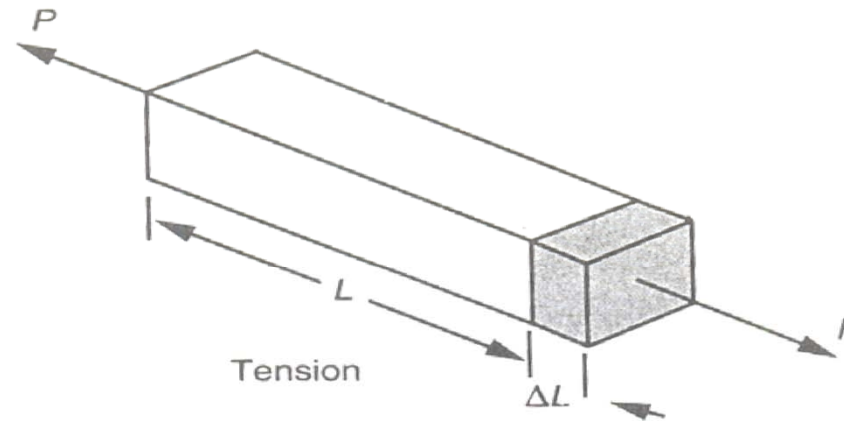
- Sizing the structure of S/C requires us to consider:
  - Stiffness
    - Measure of force required to cause a unit displacement
    - For a single-degree-of freedom system, stiffness is the inverse of flexibility
    - Structure's mode shapes and natural frequencies of vibration depend on its stiffness and mass properties
  - Strength
    - To have adequate strength, the structure must not rupture, collapse, or deform such that its function is impaired
    - Stability is structure's resistance to collapsing under compression
  - Weight
    - Designers of flight structures develop an instinct for meeting requirements with the lightest structure
- Before the design is final, we will perform many iterations that also consider:
  - Fatigue Life
  - Cost
  - Changes in Subsystem Requirements



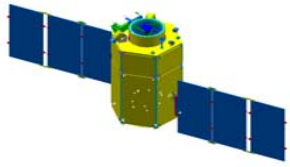
# 11.6 Structures and Mechanisms

## ◆ 11.6.6 Structural Mechanics & Analysis

- Stress:  $\sigma \equiv \text{Load/Area} = P/A \text{ N/m}^2$  or psi (pounds per square inch)
- Strain:  $\varepsilon \equiv \Delta L/L$
- Poisson's Ratio:  $\nu = \varepsilon_{\text{lateral}}/\varepsilon_{\text{axis}}$ 
  - Stiffness of a material is the relationship of its stress to strain for a given load.
- Young's Modulus (Modulus of Elasticity):  $E = \sigma/\varepsilon$



**Fig. 11-27. Stress and Strain.** Stress is load,  $P$ , divided by area,  $A$ . Strain is the change in length,  $\Delta L$ , divided by the initial length,  $L$ .



# 11.6 Structures and Mechanisms

## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

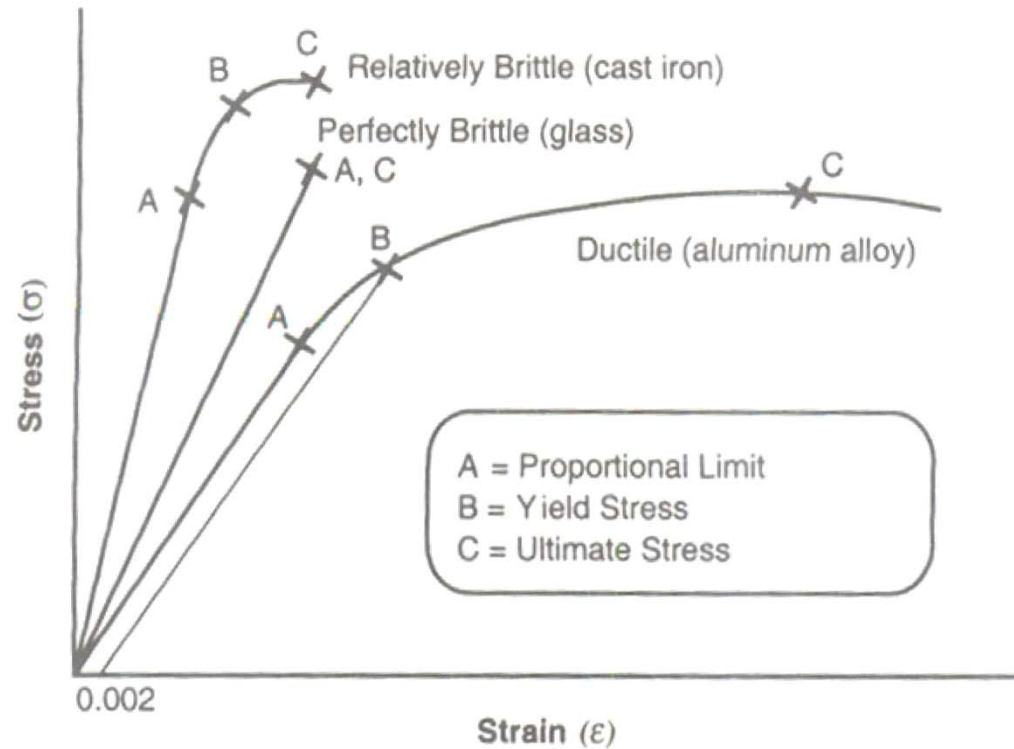
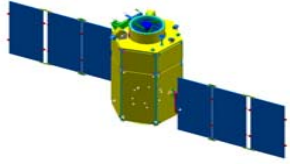


Fig. 11-28. **Representative Stress/Strain Curves for Ductile and Brittle Metals.** We generate curves such as these from uni-axial tensile tests. The slope of the linear region is the modulus of elasticity. When the material is unloaded, even when stressed above the proportional limit, stress is again proportional to strain according to the elastic modulus. The yield stress is the stress that causes a permanent strain of 0.002.

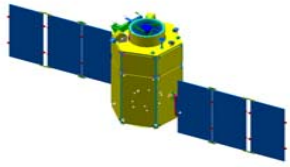


# 11.6 Structures and Mechanisms

## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

TABLE 11-55. Beam Examples.

| Name                      | Constraints                                                          | Examples                                         |
|---------------------------|----------------------------------------------------------------------|--------------------------------------------------|
| <i>Cantilevered</i>       | One end constrained against translation and rotation; other end free | Diving board                                     |
| <i>Simply Supported</i>   | Both ends constrained against translating, but free to rotate        | Plank placed across a stream for hikers to cross |
| <i>Rigidly Supported</i>  | Both ends constrained against translation and rotation               | Floor joists                                     |
| <i>Continuous Support</i> | Beam's entire span is supported                                      | Railroad track, ski                              |

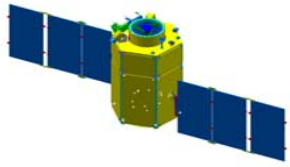


# 11.6 Structures and Mechanisms

## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

| Description                                        | Symbol                        | Schematic                                                                             | Dimension                       |
|----------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|---------------------------------|
| Concentrated axial load                            | $P$ or $F$<br>$W$ (if weight) |    | Force                           |
| Lateral, or shear load                             | $V$                           |    | Force                           |
| Concentrated reaction                              | $R$                           |    | Force<br>(equal, opposite $P$ ) |
| Uniform, distributed load                          | $w$                           |   | Force per unit length           |
| Varying, distributed load                          | $w(x)$                        |  | Force per unit length           |
| Applied bending moment,<br>reactive bending moment | $M$                           |  | Force times distance            |

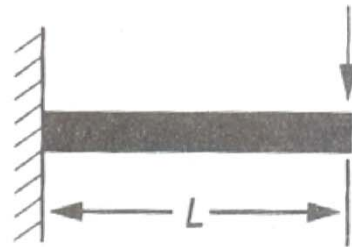
Fig. 11-29. Beam Symbols and Schematics.



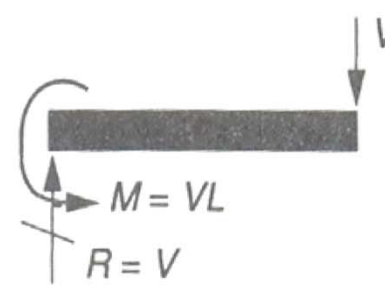
# 11.6 Structures and Mechanisms

## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

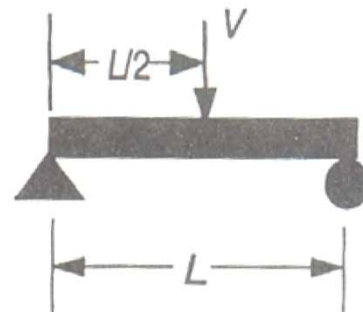
Case (A): Cantilevered



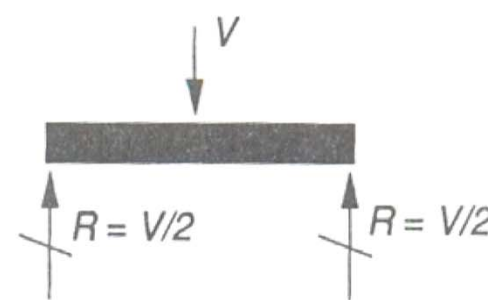
Case (A): Free-Body Diagram



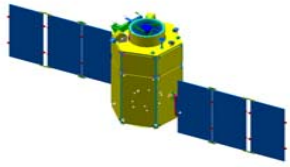
Case (B): Simply Supported



Case (B): Free-Body Diagram



**Fig. 11-30. Common Beam Cases with Associated Free-Body Diagrams.** The “brick” wall or rigid left-hand support in case (A) can be replaced with the bending moment,  $M$ , equal to  $V$  times  $L$ .



# 11.6 Structures and Mechanisms

## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

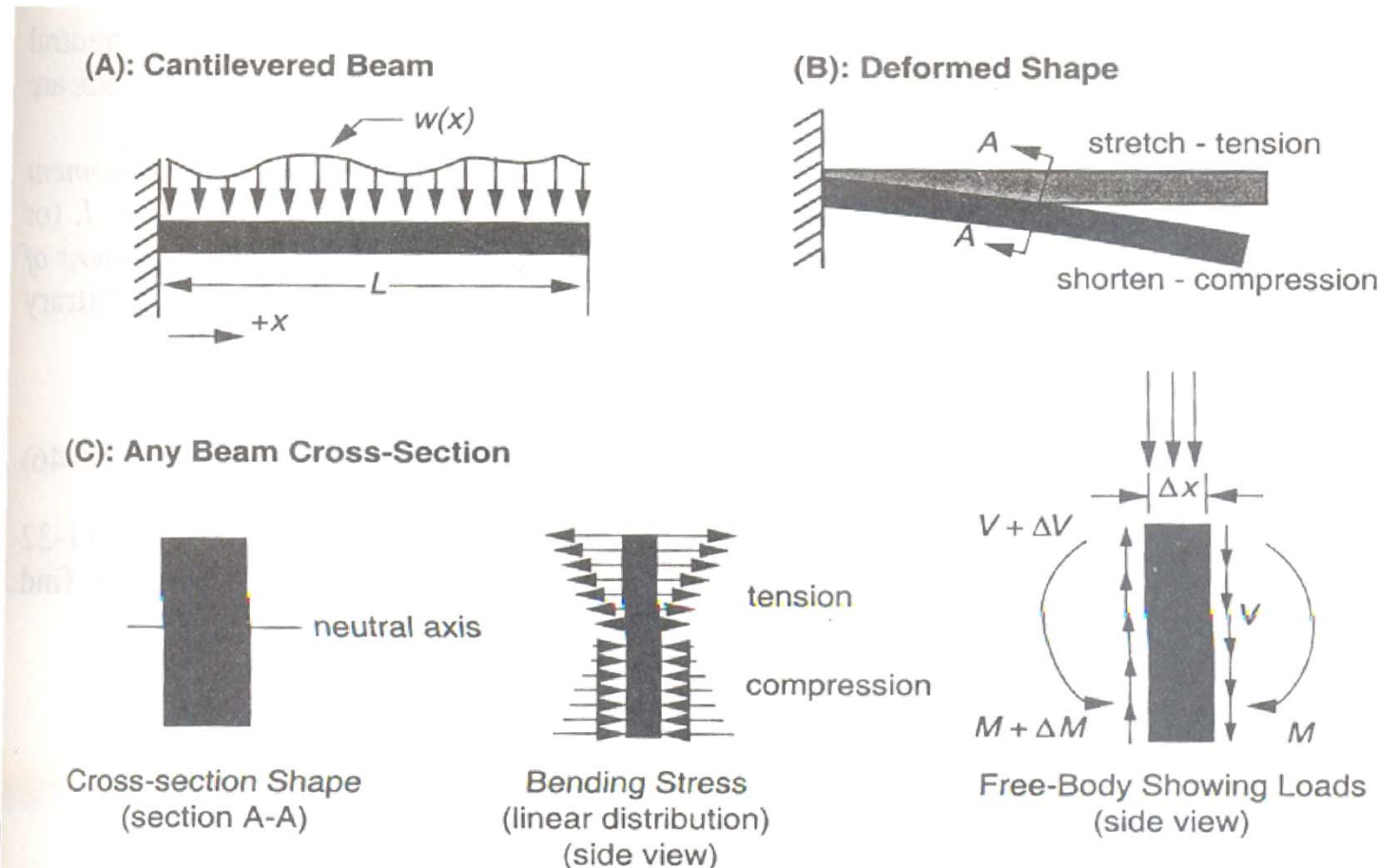
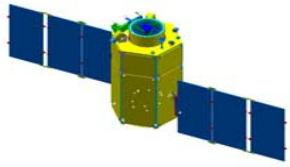


Fig. 11-31. **Bending and Shear in a Cantilevered Beam.** Bending stresses vary linearly, peaking in the parts of the section that are farthest from the *neutral axis* (centroidal bending axis). Shear stresses vary nonlinearly and are maximum at the neutral axis. Magnitudes of both bending and shear stresses vary for different cross-sections.

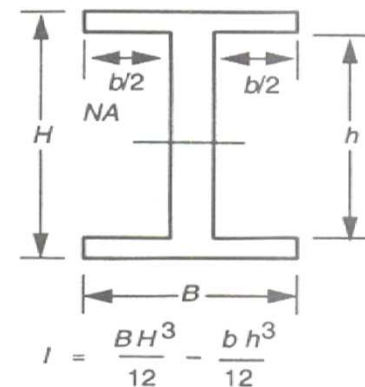
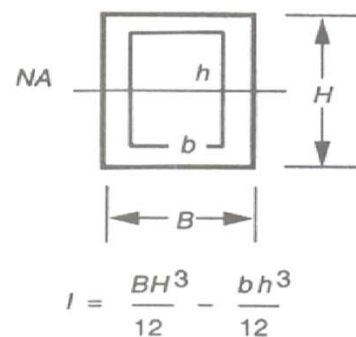
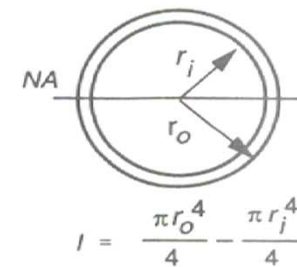
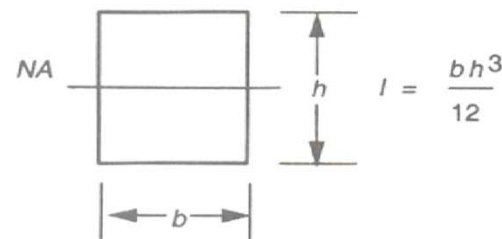


# 11.6 Structures and Mechanisms

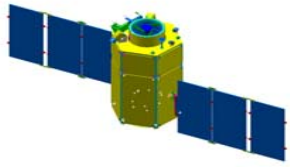
## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

- Can quantify a beam's ability to resist bending loads using area moment of inertia

- $I = \int_A y^2 dA$ ,  $y$  = distance from the centroid to  $dA$



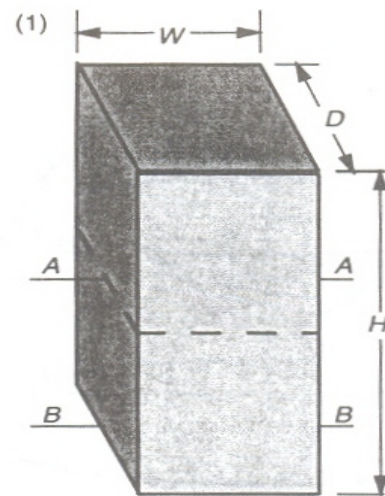
**Fig. 11-32. Common Sections and Their Centroidal Moments of Inertia.** For boxes and tubes, we find the  $I$  for the section by subtracting the inner  $I$  from the outer  $I$ . For a very thin annulus,  $I = \pi r^3 t$ .



# 11.6 Structures and Mechanisms

## 11.6.6 Structural Mechanics & Analysis (cont'd)

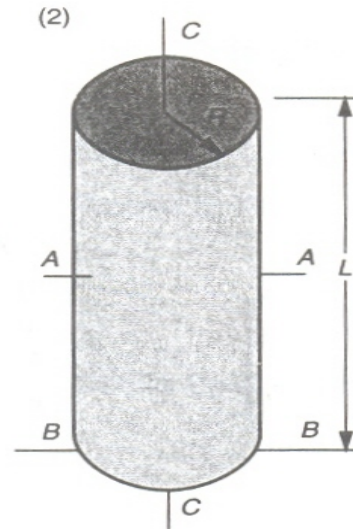
- Mass moment of inertia (MOI) is a measure of a solid's tendency to resist rotational forces.



$$m = \rho (WDH)$$

$$MOI_{AA} = \frac{m}{12} (D^2 + H^2)$$

$$MOI_{BB} = MOI_{AA} + m \left( \frac{H}{2} \right)^2$$

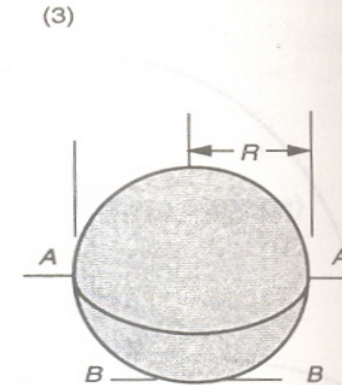


$$m = \rho (\pi R^2 L)$$

$$MOI_{AA} = \frac{m}{12} (3R^2 + L^2)$$

$$MOI_{BB} = MOI_{AA} + m \left( \frac{L}{2} \right)^2$$

$$MOI_{CC} = \frac{mR^2}{2}$$

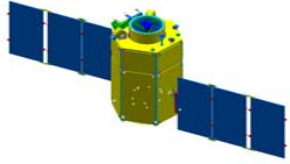


$$m = \rho \left( \frac{4\pi R^3}{3} \right)$$

$$MOI_{AA} = \frac{2}{5} mR^2$$

$$MOI_{BB} = MOI_{AA} + mR^2$$

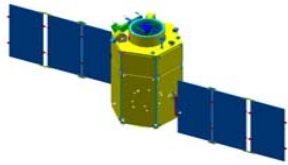
**Fig. 11-39. Equations for the Mass and Moment of Inertia for Some Common Solids.** For a solid of uniform density,  $\rho$ , the mass is the product of  $\rho$  and volume. We use the parallel axis theorem to find inertias at axes B-B.  $MOI_{BB} = MOI_{AA} + md^2$  where  $d$  is the distance from the A-A to B-B axes.



# 11.6 Structures and Mechanisms

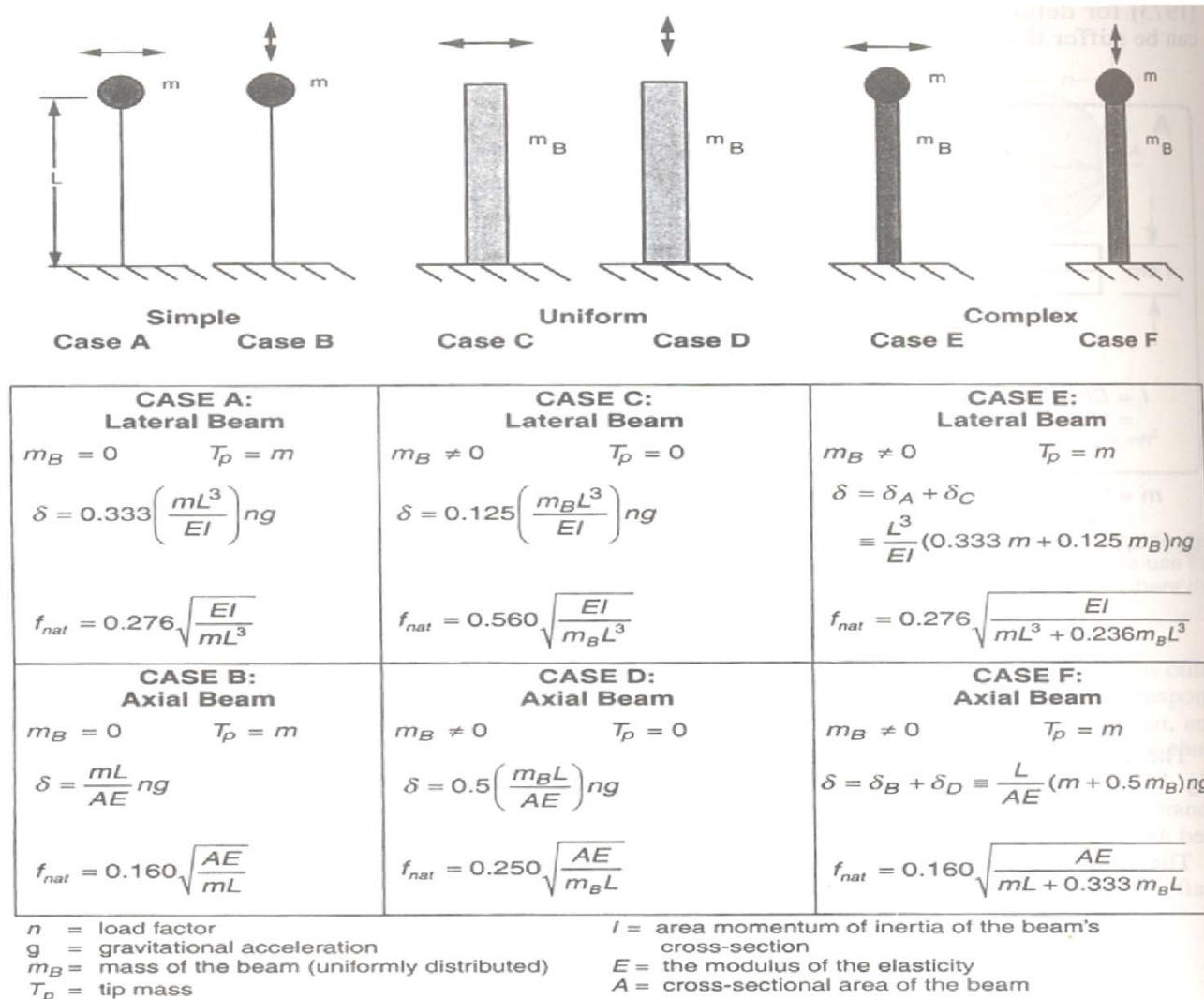
## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

- When considering only its first natural (or fundamental) frequency, a structure can be idealized as a single-degree-of-freedom spring-mass system.
- Natural Frequency:  $f_{\text{natural}} = \frac{1}{2\pi} \sqrt{k/m}$ 
  - $k = mg/\delta$ : stiffness
  - $m$ : mass
  - $\delta$ : deflection
  - $g$ : acceleration due to gravity

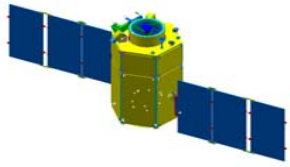


# 11.6 Structures and Mechanisms

## 11.6.6 Structural Mechanics & Analysis (cont'd)



**Fig. 11-42. Beam Deflections,  $f_{nat}$ , and Natural Frequencies,  $\delta$ .** We can estimate the natural frequencies and deflections of beams for both axial and lateral or bending loads. In cases E and F, the values of  $m$  and  $m_B$  are different from previous cases.



# 11.6 Structures and Mechanisms

## ◆ 11.6.6 Structural Mechanics & Analysis (cont'd)

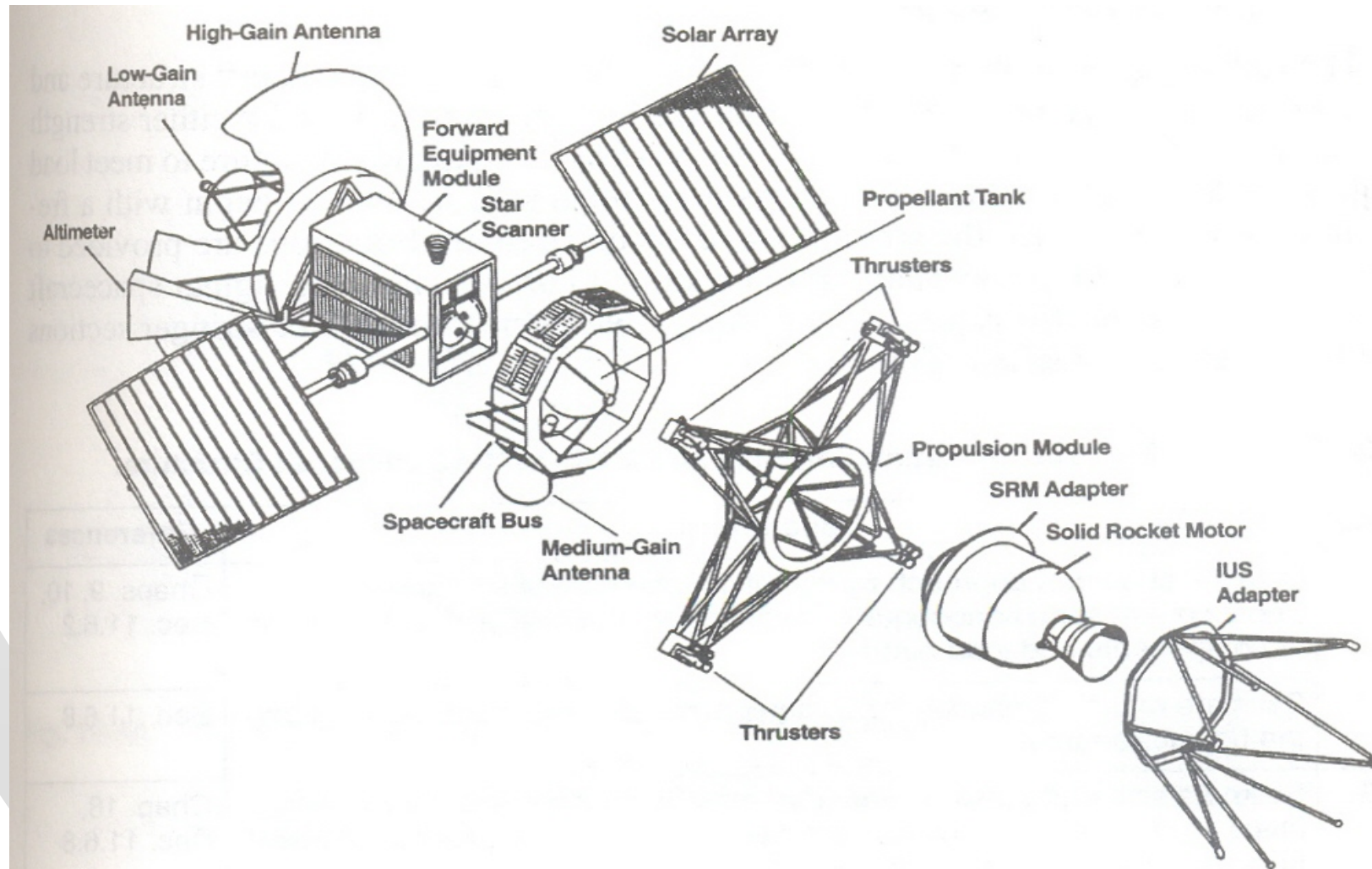
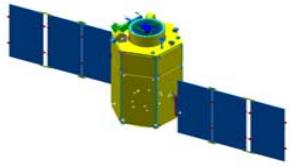


Fig. 11-43. Magellan Spacecraft.



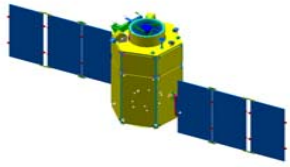
# 11.6 Structures and Mechanisms

## ◆ 11.6.7 An Example Problem

- Alternative Processes
  - Size the structure to meet load requirements and check the resulting natural frequency
  - Begin with a frequency requirement, size the structure, and check strength

**TABLE 11-56. Process for Estimating Size and Mass of the Spacecraft Structure.**

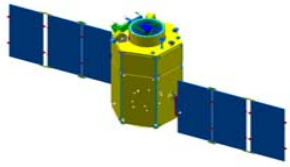
| Step | Description                                                                                                                                                                                       | References                |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1    | Select a structural approach by identifying the type of structure (monocoque, semimonocoque), shape of the structure, and arrangement of components and load paths.                               | Chaps. 9, 10, Sec. 11.6.2 |
| 2    | Estimate mass distribution for all equipment and the structure, including the booster adapter.                                                                                                    | Sec. 11.6.8               |
| 3    | Estimate size and mass of structural members using information from steps (1) and (2) and the axial and bending frequencies for the selected booster. Iterate this structural design as required. | Chap. 18, Sec. 11.6.8     |
| 4    | Apply combined design loads (axial, lateral, and bending) and determine member loads.                                                                                                             | Sec. 11.6.7               |
| 5    | Compute the structural capability and compare with the applied loads to determine the margin of safety. Iterate the design as required to obtain the necessary margin of safety.                  | Secs. 11.6.7, 11.6.8      |



# ***11.6 Structures and Mechanisms***

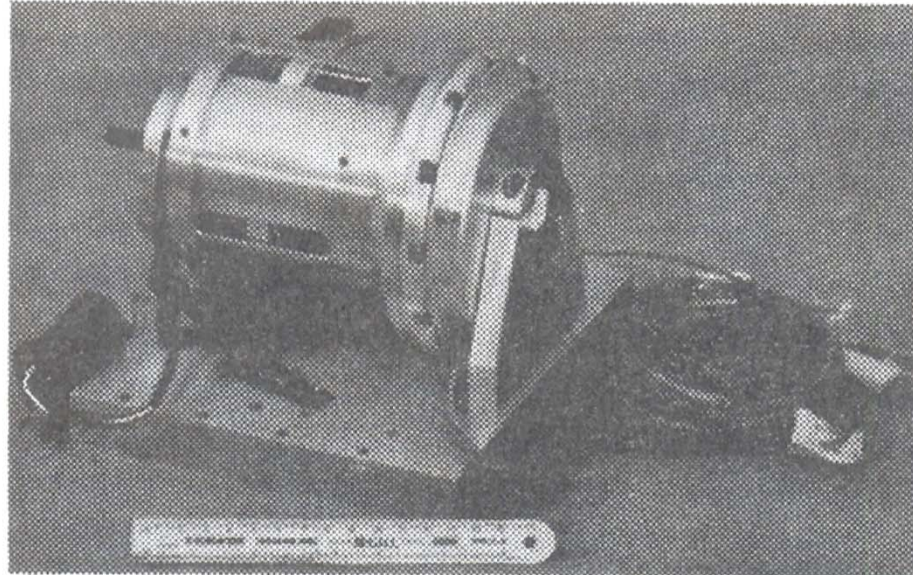
## ◆ 11.6.8 Mechanisms and Deployables

- High-Cyclic Mechanisms require frequent or constant articulation:
  - Antenna Pointing and Tracking
  - Solar Array Pointing and Tracking
  - Attitude Control Reaction Wheels
  - Boom Extensions
- Low-cyclic mechanisms:
  - Antenna Launch Retention
  - Antenna Deployment
  - Solar Array Retention
  - Solar Array Deployment
  - Spacecraft/Launch Vehicle Separation

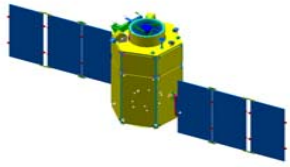


# 11.6 Structures and Mechanisms

## ◆ 11.6.8 Mechanisms and Deployables (cont'd)



**Fig. 11-46. High-Cyclic Mechanism, Rotary Actuator Assembly and Components.** An example of an aerospace mechanism requiring precision pointing (motor driven).



# 11.6 Structures and Mechanisms

## ◆ 11.6.8 Mechanisms and Deployables (cont'd)

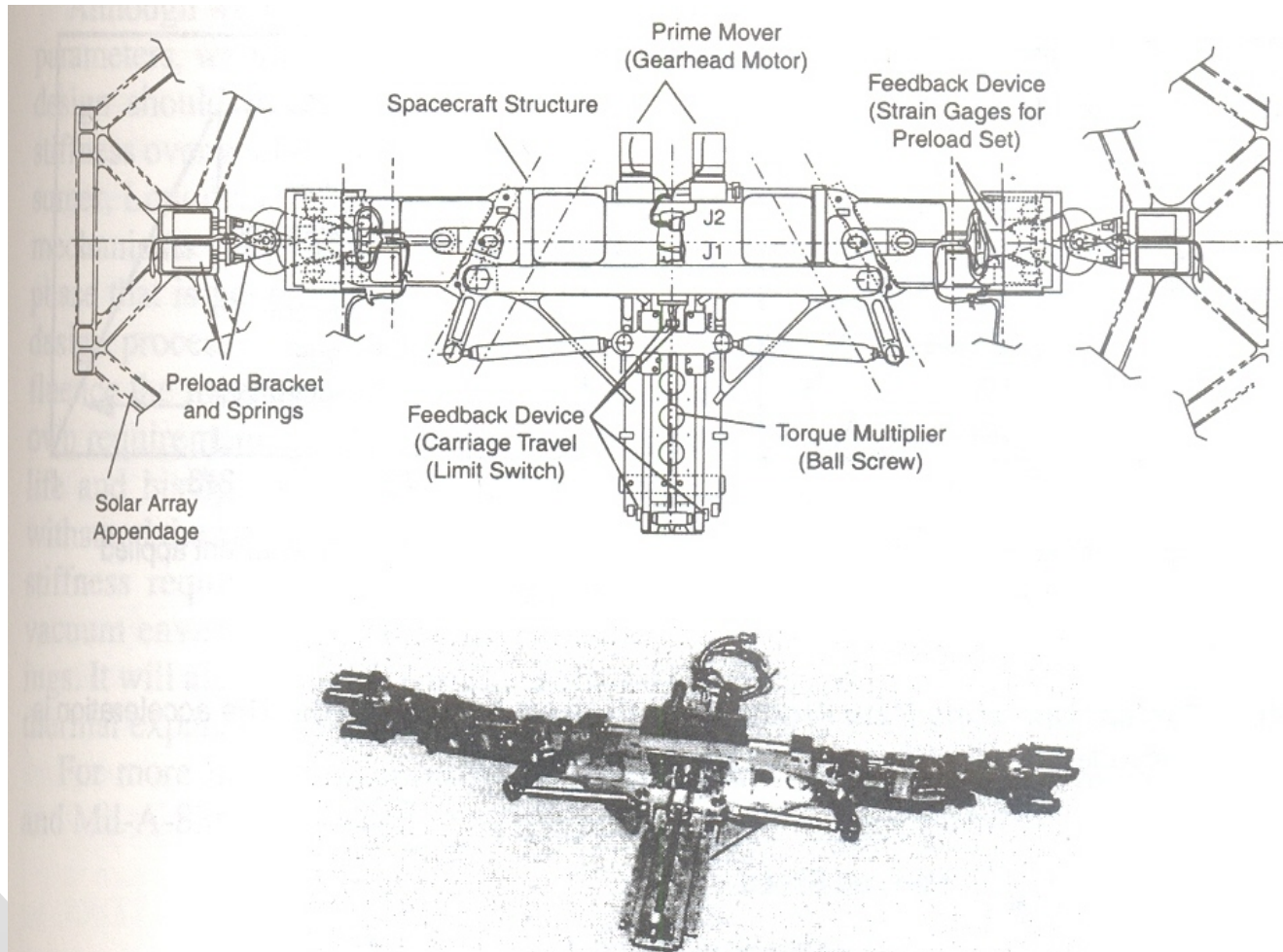


Fig. 11-47. **Low-Cyclic Mechanism, Solar Array Retention Mechanism.** An example of an aerospace appendage retention mechanism.